Questions

Q1.

(i) Figure 4 shows two light rays hitting a glass lens.

On Figure 4, draw the two light rays after they leave this lens.

(1)

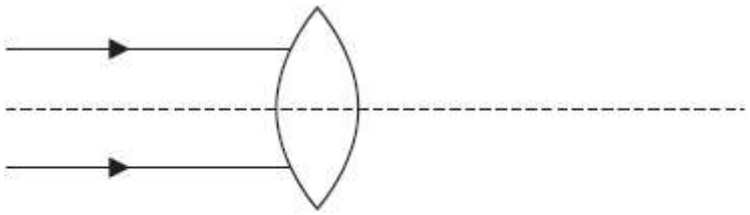

Figure 4

(ii) Figure 5 shows two light rays hitting a different glass lens.

On Figure 5, draw the two light rays after they leave this lens.

(1)

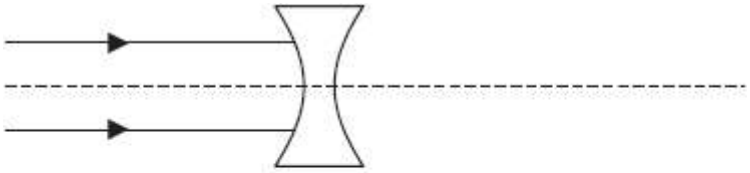

Figure 5

(iii) A lens has a focal length of 25 cm.

Calculate the power of the lens.
Use the equation

$$\text{power in dioptres} = \frac{1}{\text{focal length in metres}}$$

(2)

power of the lens = ... dioptres

(Total for question = 4 marks)

Q2.

Figure 1 shows a ray of light incident on a mirror.

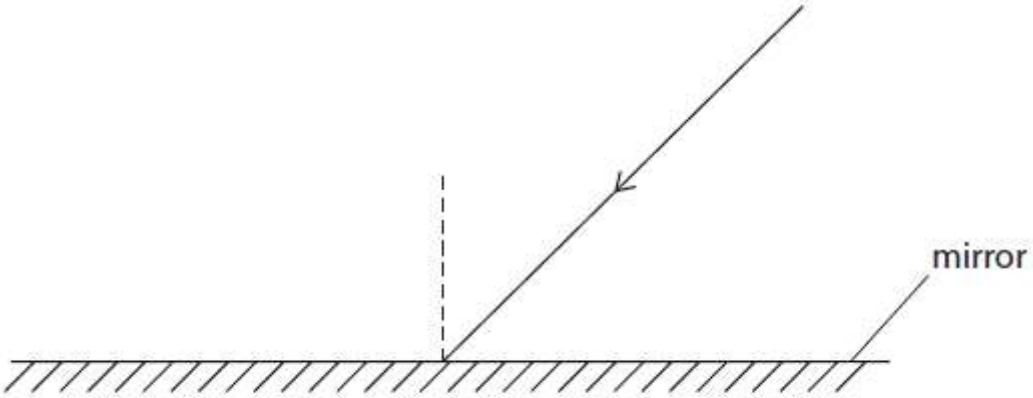

Figure 1

(i) Draw the reflected ray on Figure 1.

(1)

(ii) On Figure 1, label the angle of incidence with the letter **i** and the angle of reflection with the letter **r**.

(2)

(Total for question = 3 marks)

Q3.

This question is about reflection and refraction of light.

(i) Figure 4 shows a ray of light travelling to a plane mirror.
On Figure 4, draw the ray of light after it **reflects** off the mirror surface.

(2)

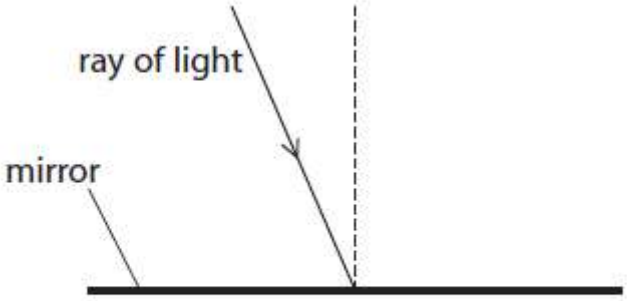

Figure 4

(ii) Figure 5 shows a ray of light in air travelling to a glass block.
On Figure 5, draw the ray of light after it **refracts** at the surface of the glass block.

(2)

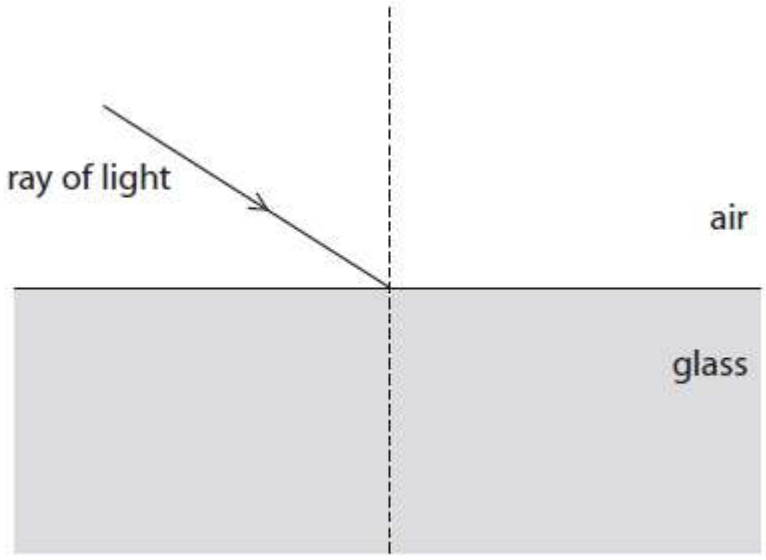

Figure 5

(iii) Figure 6 shows a ray of light in water, travelling to the surface of the water.
The angle marked **X** is greater than the critical angle.
On Figure 6, draw the ray of light after it reaches the surface of the water.

(2)

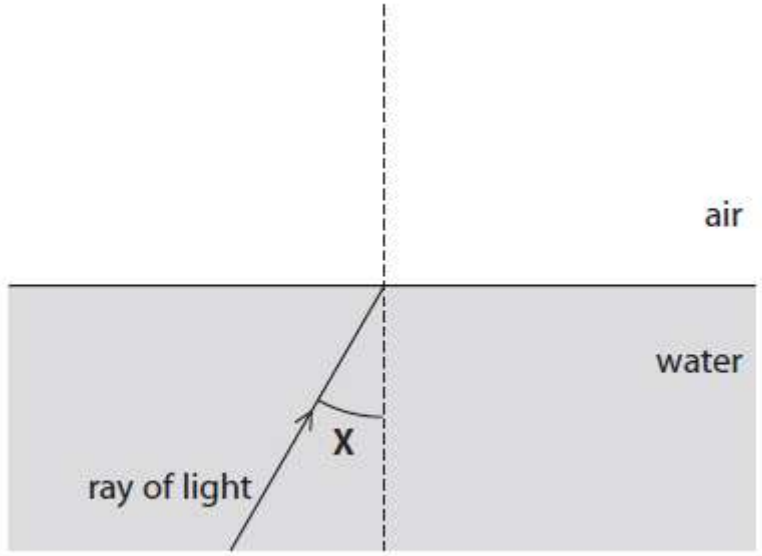

Figure 6

(Total for question = 6 marks)

Q4.

Draw **one** straight line from each electromagnetic radiation to its correct description.

(3)

electromagnetic radiation	description
infrared	used in night vision cameras
gamma	the least dangerous radiation
ultraviolet	may damage eyes, so sunglasses worn as protection
radio waves	the most dangerous radiation

(Total for question = 3 marks)

Q5.

Draw **one** line from each **use of wave** to the matching **electromagnetic wave**.

One line has been drawn for you.

(3)

use of wave	electromagnetic wave
to detect forged banknotes	radio waves
	microwaves
to detect broken bones	infrared waves
	visible light
for night-vision cameras	ultraviolet waves
	X-rays
to sterilise medical equipment	gamma rays

(Total for question = 3 marks)

Q6.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question is about waves in the electromagnetic (e-m) spectrum.

(i) Figure 1 shows some types of radiation that form part of the e-m spectrum and some uses of e-m radiation.

Draw **one** straight line from each type of e-m radiation to its use.
One line has been drawn for you.

(3)

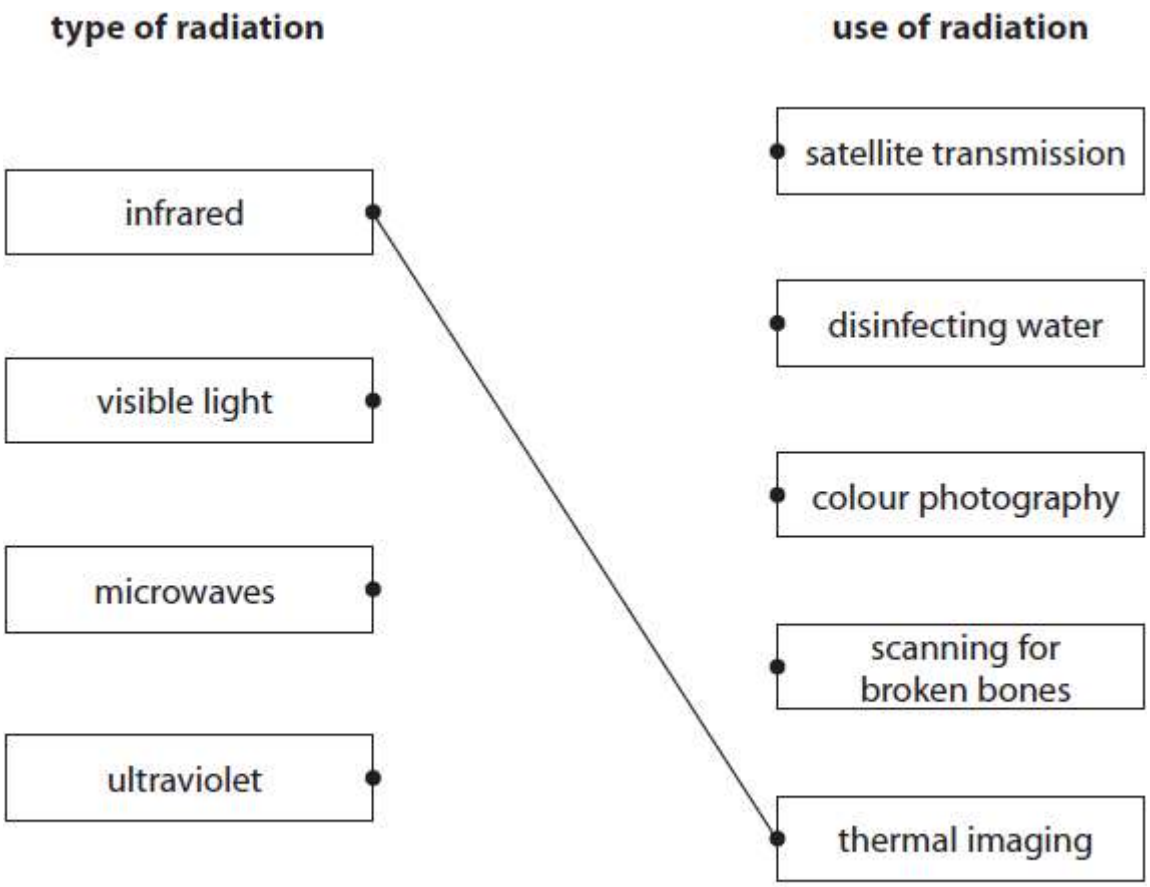

Figure 1

(ii) Which of these waves has the highest frequency?

(1)

- ☐ **A** infrared
- ☐ **B** microwaves
- ☐ **C** ultraviolet
- ☐ **D** visible light

(Total for question = 4 marks)

Q7.

Figure 12 shows ray diagrams with objects placed at different distances from lenses.

(3)

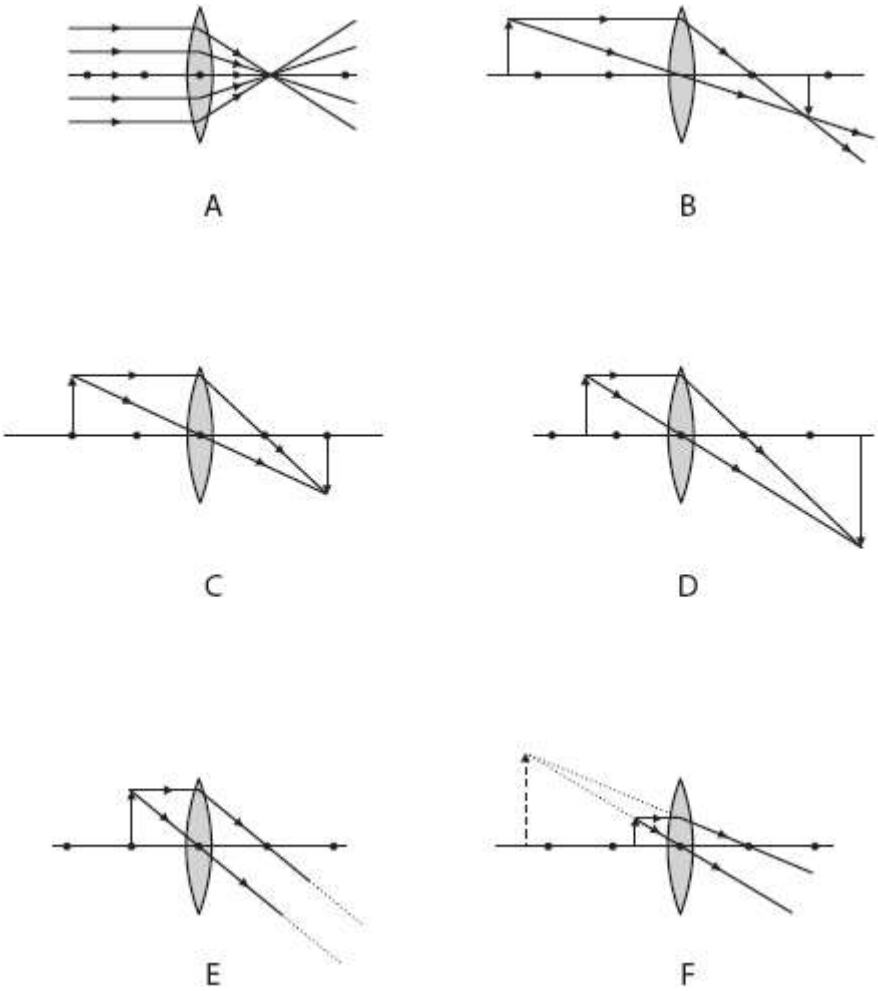

Figure 12

A student writes the conclusion:

'*All lenses produce real images*'.

Comment on the student's conclusion.

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(Total for question = 3 marks)

Q8.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Figure 6 shows two solid metal balls, **P** and **Q**.

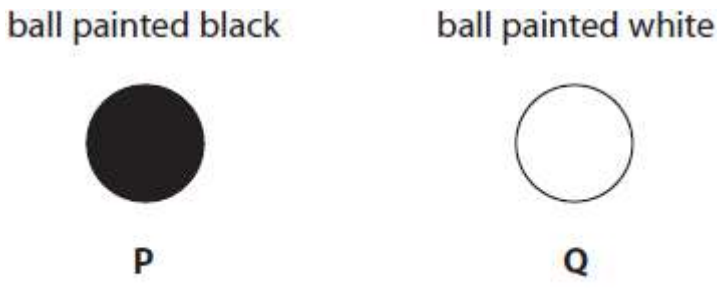

Figure 6

P and **Q** are made from the same metal and have the same radius.

P is painted black and **Q** is painted white.

Each ball is heated to a different temperature.

The balls then cool in the same room.

The graph in Figure 7 shows how the temperature of each ball changes with time.

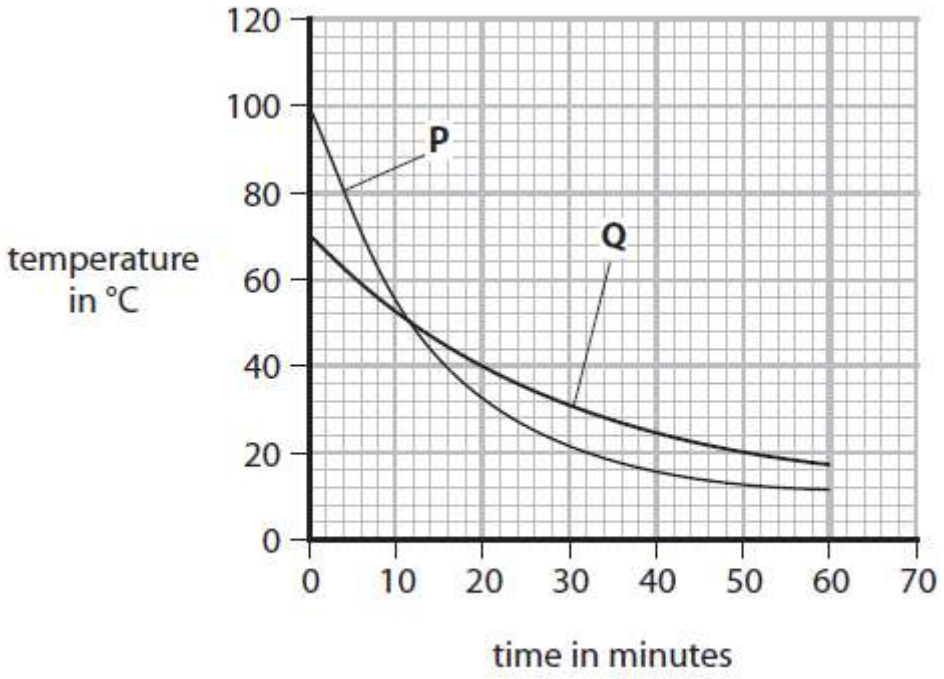

Figure 7

(i) Use the graph in Figure 7 to determine the time when **P** and **Q** were at the same temperature.
Show your working on the graph.

(2)

time = ... minutes

(ii) Which of these temperatures is most likely to be room temperature, as shown by the graph in Figure 7?

(1)

☐ **A** 100 °C
☐ **B** 70 °C
☐ **C** 10 °C
☐ **D** 0 °C

(iii) Explain why the curve for **P** is different from the curve of **Q**. Use information from Figure 6 and Figure 7 to
help your answer.

(2)

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(Total for question = 5 marks)

Q9.

A student investigates how the surface of an object affects the radiation it emits.

Figure 7 shows the equipment he uses.

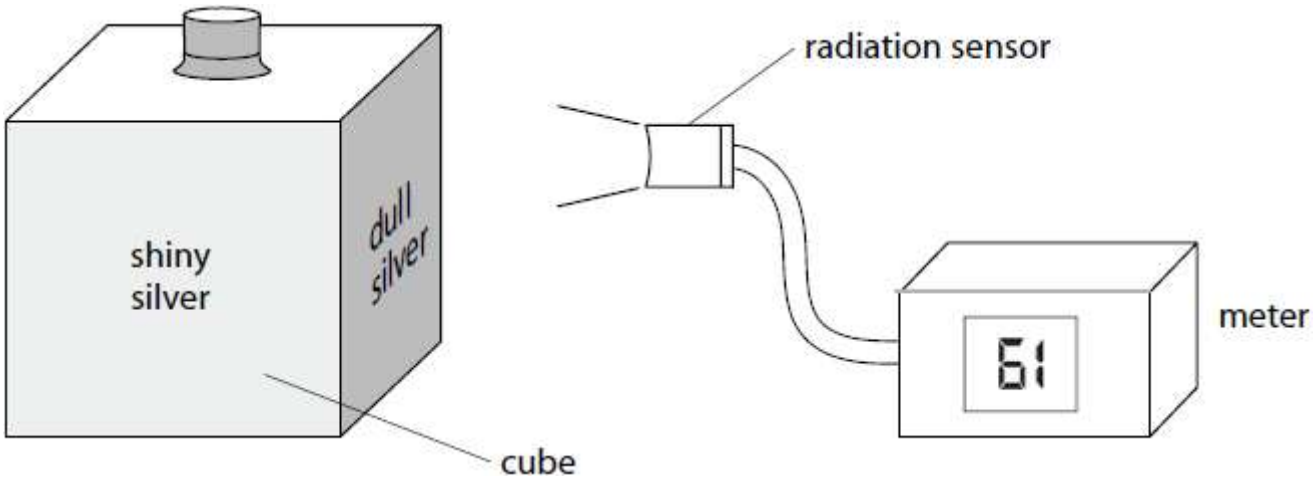


Figure 7

The cube has four different surfaces.

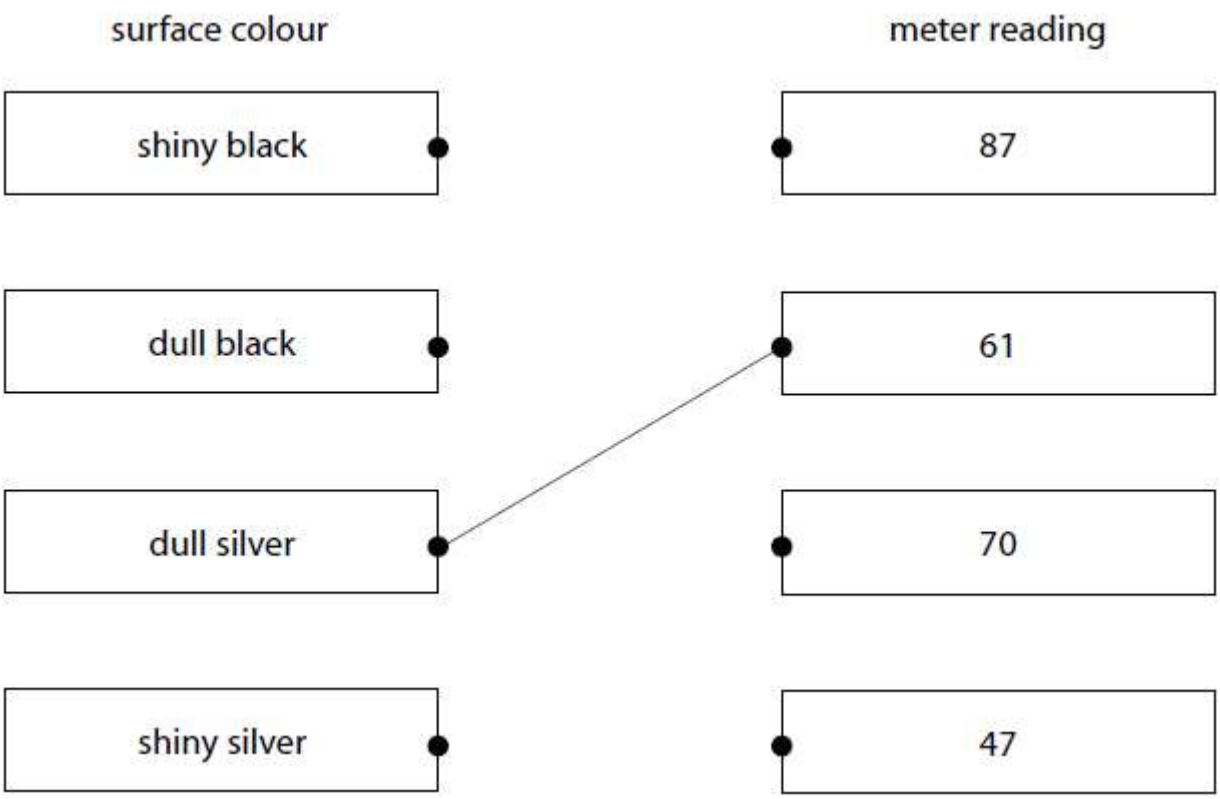
He fills the cube with boiling water so that the temperature of each surface is the same.

He uses the radiation sensor to measure the radiation emitted from each surface.

(i) His readings are shown.

Draw a line from each surface colour to its correct meter reading. One has been done for you.

(2)



(ii) The temperature of each surface is the same.

Give a reason why the radiation sensor gives a different reading for each surface.

(1)

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(Total for question = 3 marks)

Q10.

A student investigates how light behaves as it leaves a clear plastic block.

Figure 4 shows some of her equipment and the path of a ray of light through the block.

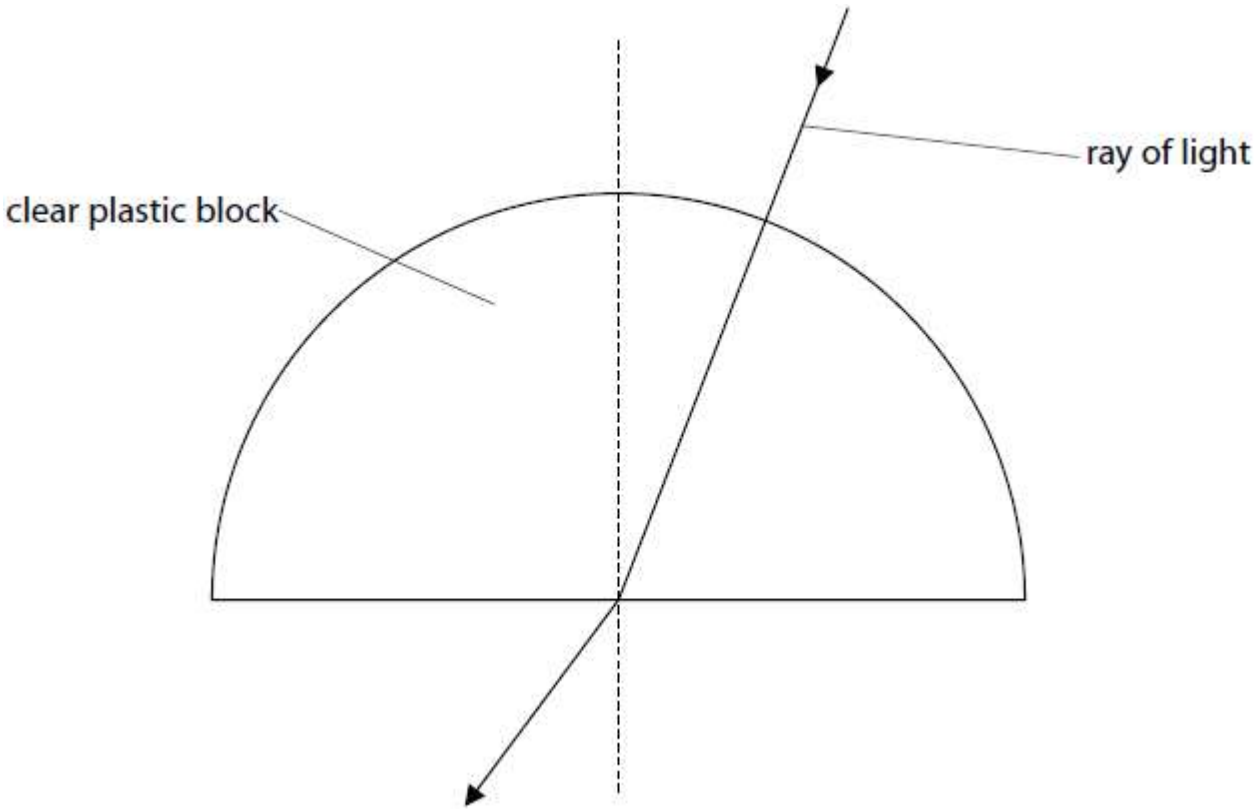

Figure 4

(a) Describe how the student can make sure the light does not change direction as it enters the block.

(2)

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(b) Describe what happens to the ray of light when the angle of incidence is increased beyond the critical angle.

(2)

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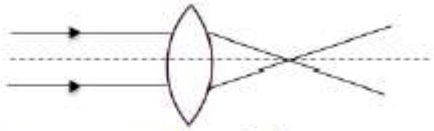
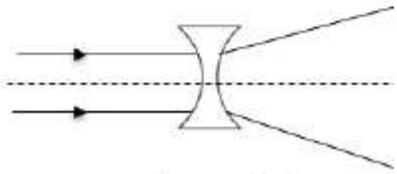
(Total for question = 4 marks)

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
(i)	rays converging	 arrows not needed	(1)

Question number	Answer	Additional guidance	Mark
(ii)	rays diverging	 arrows not needed award 1mark if convergence and divergence are shown but with the wrong lenses	(1)

Question number	Answer	Additional guidance	Mark
(iii)	substitution (1) $\frac{1}{25(\times 10^{-2})}$ evaluation (1) 4(.0)	or 0.04 seen ignore powers of ten until evaluation award full marks for the correct answer without working	(2)

Q2.

Q3.

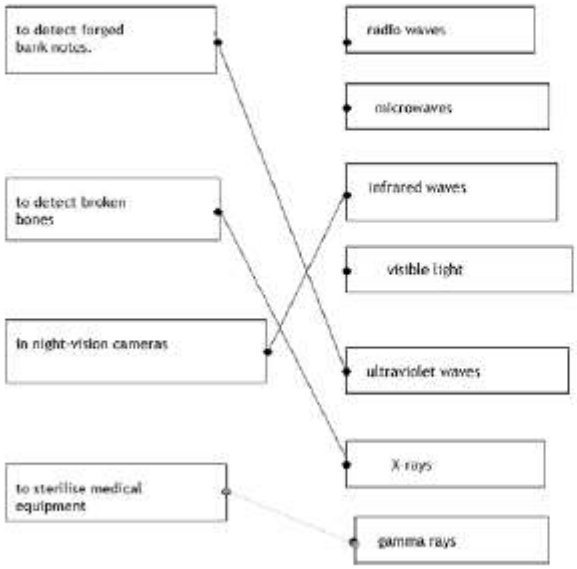
	Answer	Additional guidance	Mark
(i)	diagram to include: a re**flect**ed ray drawn (1) angle of reflection = angle of incidence (1)	 judge by eye	(2) AO2

	Answer	Additional guidance	Mark
(ii)	diagram to include: a re**fract**ed ray drawn (1) angle of refraction < angle of incidence (1)	 Ray drawn in bottom right quadrant of diagram ignore reflected rays judge by eye	(2) AO2

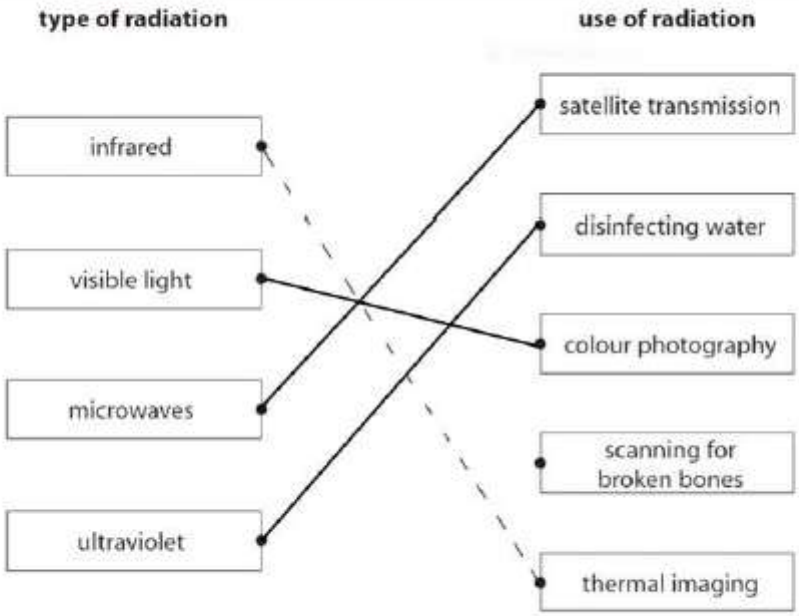
	Answer	Additional guidance	Mark
(iii)	diagram to include: ray drawn showing total internal reflection (1) angle of reflection = angle of incidence (1)	 REJECT any refracted ray for this mark judge by eye	(2) AO2

Q4.

Q5.

Question number	Answer	Additional guidance	Mark
		award 1 mark for each line from the three left-hand boxes more than one line from a box loses the mark for that box	(3)

Q6.

Question Number	Answer	Additional guidance	Mark
(i)		award one mark for each correct line up to three marks reject for a mark two lines starting or ending at the same box	(3) AO1

Question Number	Answer	Mark
(ii)	C ultraviolet A (infrared), B (microwaves) and D (visible light) all have frequencies below that of ultraviolet	(1) AO1

Q7.

Question number	Answer	Additional guidance	Mark
	A comment that makes reference to the following points: • Diagram F shows a virtual image (1) • Not all lenses are convex (1) and therefore the student's conclusion is wrong (1)	Last marking point can only be achieved if at least one of the other two marks id awarded	(3)

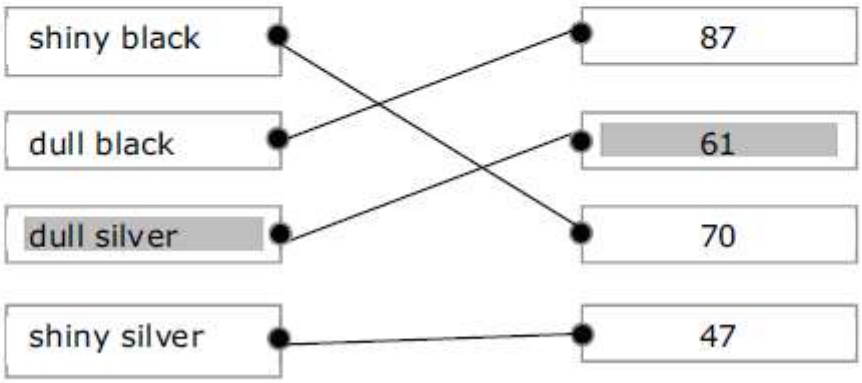
Q8.

Question number	Answer	Additional guidance	Mark
(i)	line shown on graph from intersection of two curves (1) answer in range 11 – 13 (minutes) inclusive (1)	award full marks for the correct answer without working	(2)

Question number	Answer	Mark
(ii)	☒ **C** 10 °C	(1)

Question number	Answer	Additional guidance	Mark
(iii)	an explanation linking: the gradient for **P** is greater/steeper than the gradient for **Q** (1) (because) **P**/black is a better emitter (of radiation) (than **Q**/white) (1)	 **P** cools quicker than **Q** **P** loses thermal energy/heat quicker than **Q** allow reverse arguments credit answers in terms of absorption in this context	(2)

Q9.

Question number	Answer	Mark
(i)	All three correct (2) One or two correct (1)  	(2)

Question number	Answer	Additional guidance	Mark
(ii)	Different surfaces emit (thermal) radiation at different rates	allow reference to surfaces in question	(1)

Q10.

Question number	Answer	Additional guidance	Mark
(a)	An answer that combines the following points of understanding to provide a logical description: • shine the light along a radius (1) • by marking it on the paper before putting the block down (1)	allow shine the ray at the centre of the straight edge before putting the block down	(2)

Question number	Answer	Mark
(b)	An answer that provides a description by making reference to: • (all) light reflected (1) • back inside block (1)	(2)